COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE010	**SECTION:**	CPE21S4
DATE PERFORMED:	07/29/2026	**DATE SUBMITTED:**	07/29/2026
NAME:	KEITH NICOLE SILVOZA	**INSTRUCTOR:**	JIMLORD QUEJADO

SW 4.1

Array-Based vs. STL Stack Implementations

1. PROCEDURE OUTPUT

What procedures did you perform?

Objective

- Implement a Stack Data Structure using two distinct approaches in C++: a custom Static Array implementation and the C++ Standard Template Library

Tasks & Intended Learning Outcomes (ILOs):

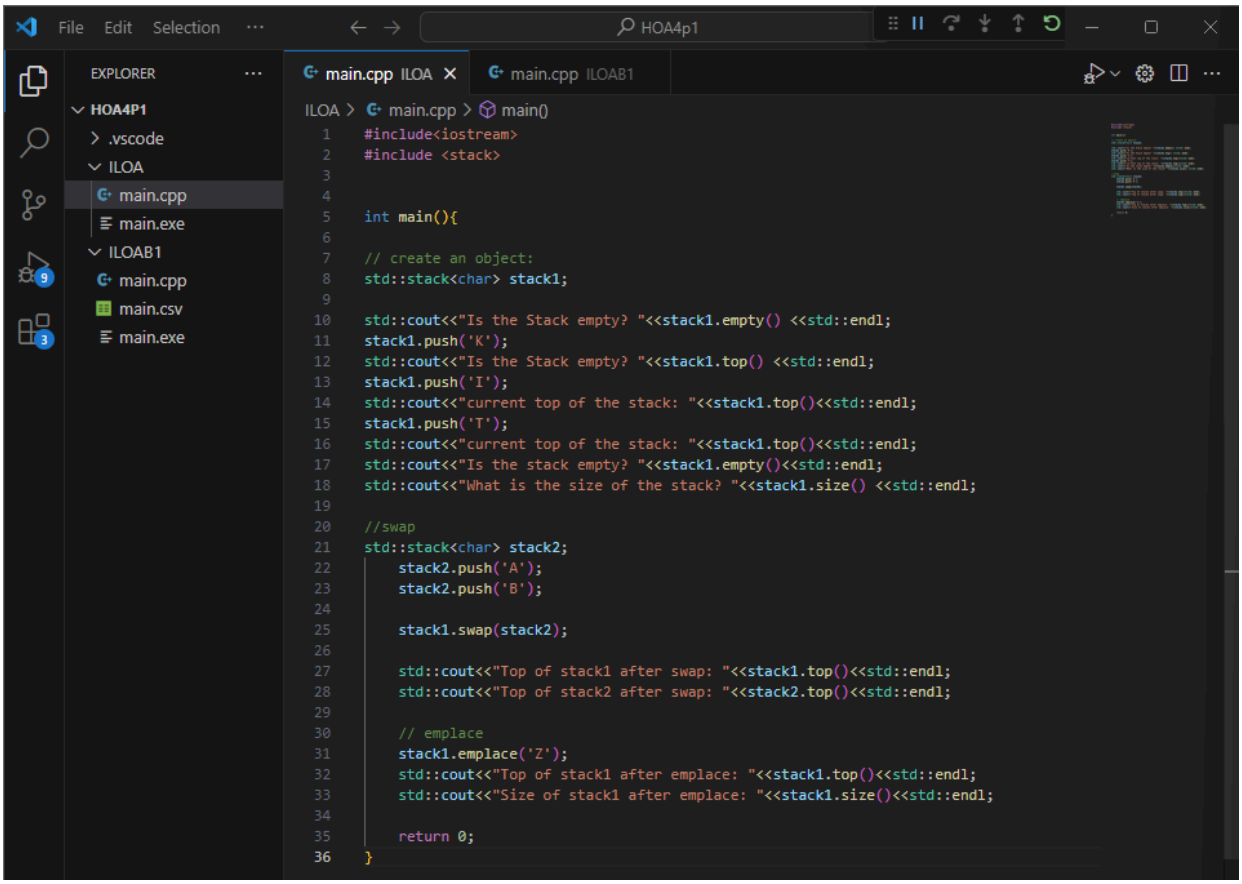
- **ILO A:** Implement a stack using the C++ STL (`std::stack`).
- **ILO B.1:** Implement a stack using a Static Array.

Submission Requirements:

- **Documentation:** Please use the official **Activity Template** for your write-up.
- **Source Code:** Submit your C++ file named as `LastName_Seatwork4_1.cpp`.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

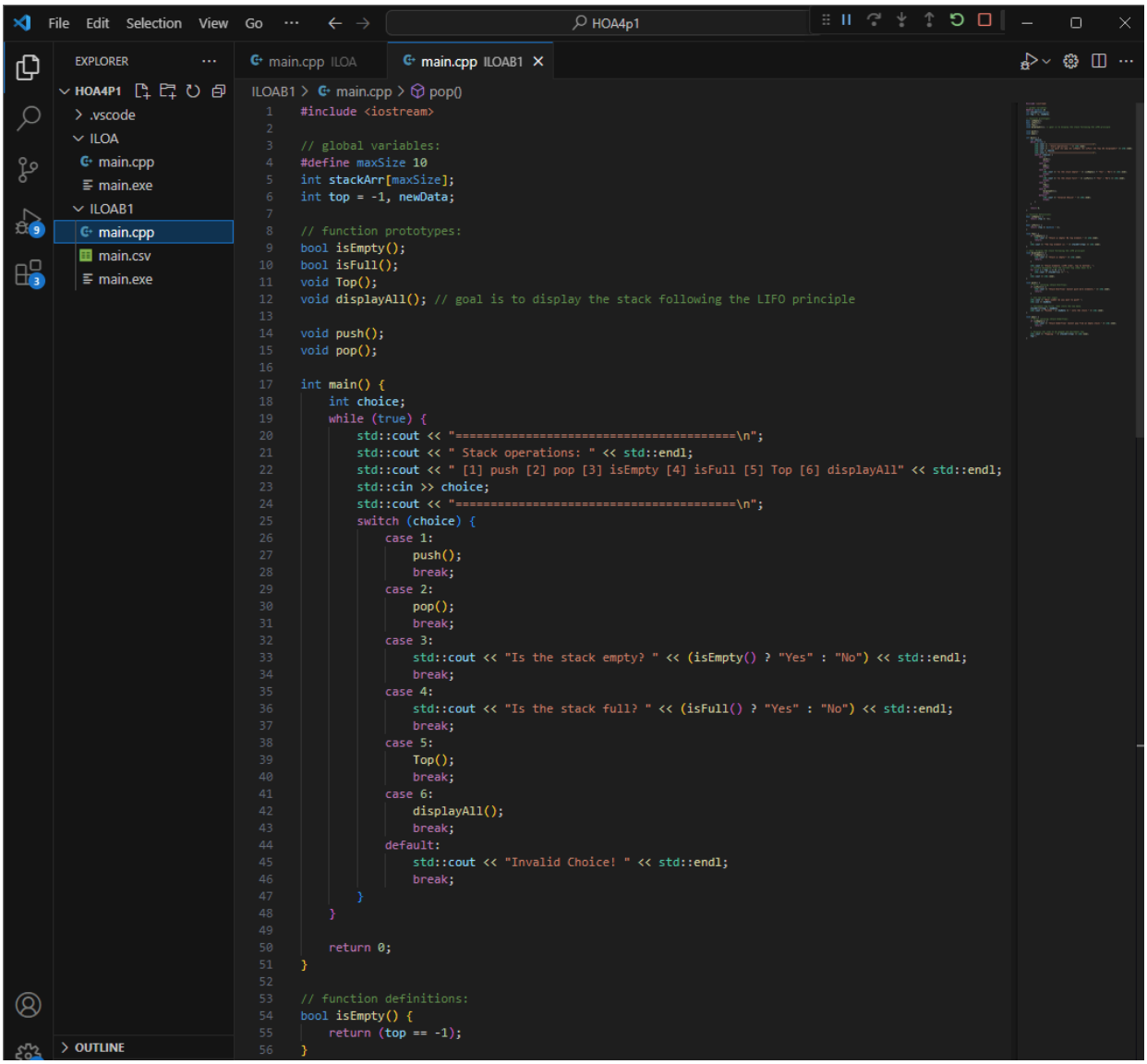
ILOA



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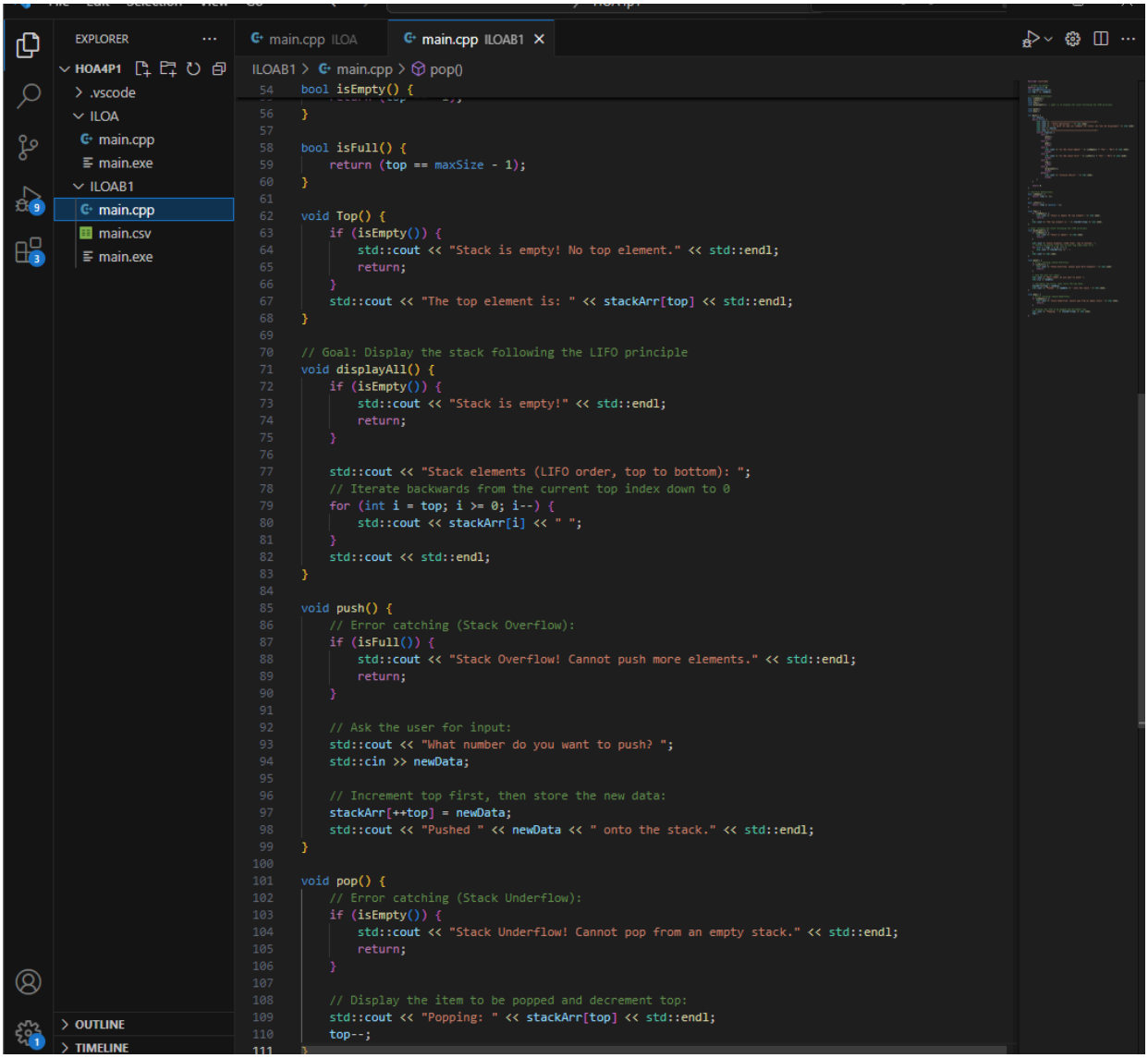
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ILOAB1

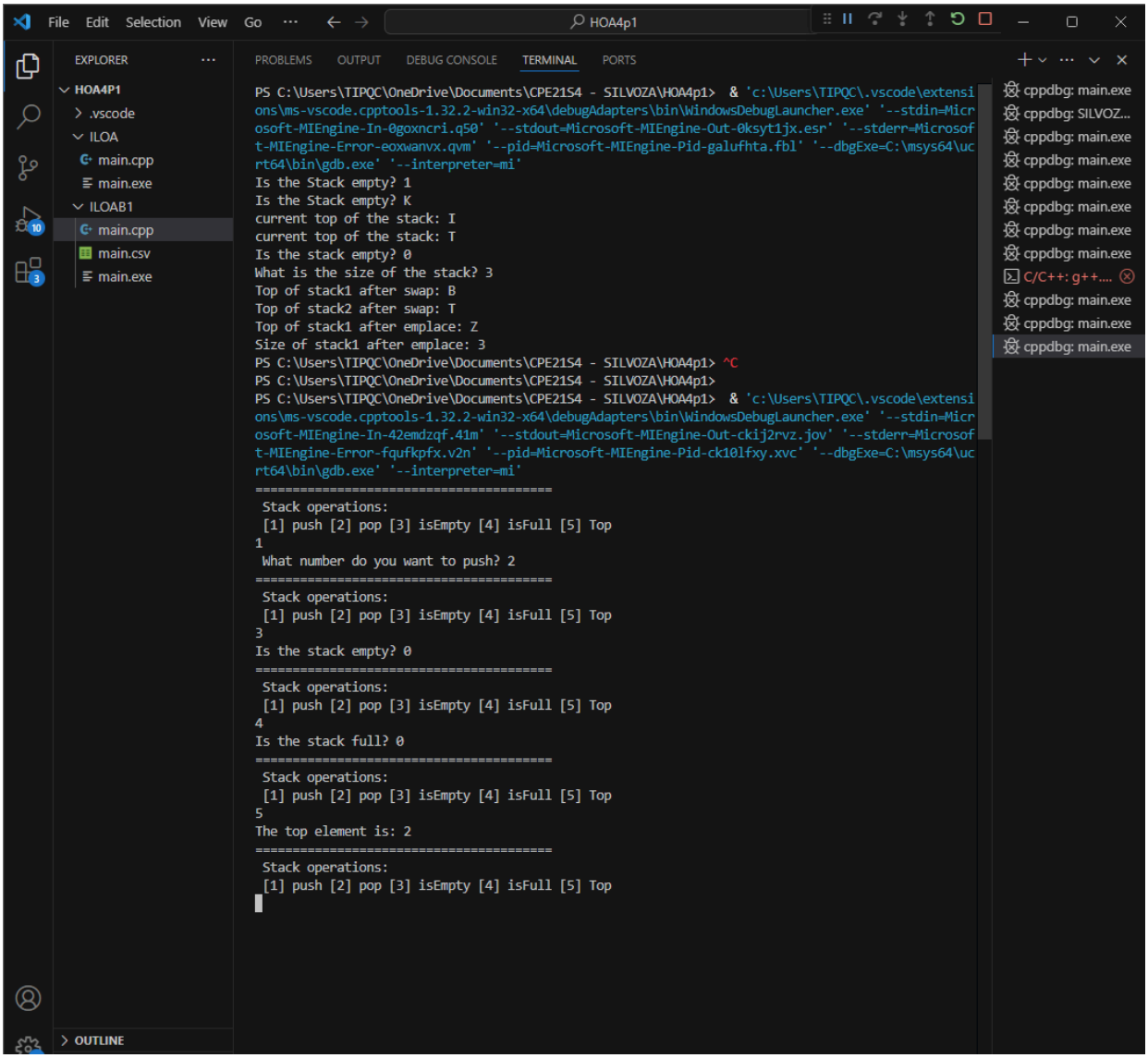


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OUTPUT FOR ILOA & ILOAB1



ANALYSIS

- The program uses an array-based stack.
- It includes Push, Pop, Empty, Full, and Display operations.
- The stack follows the LIFO (Last In, First Out) principle.
- The output shows that the stack operations work correctly.
- The stack's top element and size are updated after each operation.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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